

LOC	OBJECT CODE	ADDR1	ADDR2	STMT
				2 *****
				3 *
				4 *
				5 * Miscellaneous-instruction-extensions facility 4 instruction
				6 * tests for RRF-a encoded:
				7 *
				8 * B96C BEXTG - Bit Extract
				9 * B96D BDEPG - Bit Deposit
				10 *
				11 * James Wekel March 2026
				12 *****
				14 *****
				15 *
				16 * basic instruction tests
				17 *
				18 *****
				19 * This program tests proper functioning of the Miscellaneous
				20 * Instruction Extensions facility 4 RRF-a Bit Depost and Bit Extract
				21 * instructions.
				22 * Exceptions are not tested.
				23 *
				24 * PLEASE NOTE that the tests are very SIMPLE TESTS designed to catch
				25 * obvious coding errors. None of the tests are thorough. They are
				26 * NOT designed to test all aspects of any of the instructions.
				27 *
				28 *****
				29 *
				30 * *Testcase mie4-BDEPG-BDEPG
				31 * *
				32 * * Miscellaneous-instruction-extensions facility 4 instruction
				33 * * tests for RRF-a encoded:
				34 * *
				35 * * B96C BEXTG - Bit Extract
				36 * * B96D BDEPG - Bit Deposit
				37 * *
				38 * * # -----
				39 * * # This tests only the basic function of the instruction.
				40 * * # Exceptions are NOT tested.
				41 * * # -----
				42 * *
				43 * mainsize 2
				44 * numcpu 1
				45 * sysclear
				46 * archlvl z/Arch
				47 *
				48 * loadcore "\$(testpath)/mie4-BDEPG-BDEPG.core" 0x0
				49 *
				50 * diag8cmd enable # (needed for messages to Hercules console)
				51 * runtest 5 #
				52 * diag8cmd disable # (reset back to default)
				53 *
				54 * *Done
				55 *
				56 *

LOC	OBJECT CODE	ADDR1	ADDR2	STMT
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57 *****

LOC	OBJECT CODE	ADDR1	ADDR2	STMT
				59 *****
				60 * FCHECK Macro - Is a Facility Bit set?
				61 *
				62 * If the facility bit is NOT set, an message is issued and
				63 * the test is skipped.
				64 *
				65 * Fcheck uses R0, R1 and R2
				66 *
				67 * eg. FCHECK 134,'vector-packed-decimal'
				68 *****
				69 MACRO
				70 FCHECK &BITNO,&NOTSETMSG
				71 .* &BITNO : facility bit number to check
				72 .* &NOTSETMSG : 'facility name'
				73 LCLA &FBBYTE Facility bit in Byte
				74 LCLA &FBBIT Facility bit within Byte
				75
				76 LCLA &L(8)
				77 &L(1) SetA 128,64,32,16,8,4,2,1 bit positions within byte
				78
				79 &FBBYTE SETA &BITNO/8
				80 &FBBIT SETA &L((&BITNO-(&FBBYTE*8))+1)
				81 .* MNOTE 0,'checking Bit=&BITNO: FBBYTE=&FBBYTE, FBBIT=&FBBIT'
				82
				83 B X&SYSNDX
				84 * Fcheck data area
				85 * skip messgae
				86 SKT&SYSNDX DC C' Skipping tests: '
				87 DC C&NOTSETMSG
				88 DC C' (bit &BITNO) is not installed.'
				89 SKL&SYSNDX EQU *-SKT&SYSNDX
				90 * facility bits
				91 DS FD gap
				92 FB&SYSNDX DS 4FD
				93 DS FD gap
				94 *
				95 X&SYSNDX EQU *
				96 LA R0,((X&SYSNDX-FB&SYSNDX)/8)-1
				97 STFLE FB&SYSNDX get facility bits
				98
				99 XGR R0,R0
				100 IC R0,FB&SYSNDX+&FBBYTE get fbit byte
				101 N R0,=F'&FBBIT' is bit set?
				102 BNZ XC&SYSNDX
				103 *
				104 * facility bit not set, issue message and exit
				105 *
				106 LA R0,SKL&SYSNDX message length
				107 LA R1,SKT&SYSNDX message address
				108 BAL R2,MSG
				109
				110 B EOJ
				111 XC&SYSNDX EQU *
				112 MEND

LOC	OBJECT CODE	ADDR1	ADDR2	STMT	
				150	*****
				151	* Low core PSWs
				152	*****
00000000		00000000	00001C1F	153	MIE4TST START 0
		00000000		154	USING MIE4TST,R0 Low core addressability
		00000140	00000000	155	
				156	SVOLDPSW EQU MIE4TST+X'140' z/Arch Supervisor call old PSW
00000000		00000000	000001A0	158	ORG MIE4TST+X'1A0' z/Architecure RESTART PSW
000001A0	00000001 80000000			159	DC X'0000000180000000'
000001A8	00000000 00000200			160	DC AD(BEGIN)
000001B0		000001B0	000001D0	162	ORG MIE4TST+X'1D0' z/Architecure PROGRAM CHECK PSW
000001D0	00020001 80000000			163	DC X'0002000180000000'
000001D8	00000000 0000DEAD			164	DC AD(X'DEAD')
000001E0		000001E0	00000200	166	ORG MIE4TST+X'200' Start of actual test program...
				168	*****
				169	* The actual "MIE4TST" program itself...
				170	*****
				171	*
				172	* Architecture Mode: z/Arch
				173	* Register Usage:
				174	*
				175	* R0 (work)
				176	* R1-4 (work)
				177	* R5 Testing control table - current test base
				178	* R6-R7 (work)
				179	* R8 First base register
				180	* R9 Second base register
				181	* R10 Third base register
				182	* R11 M4TEST call return
				183	* R12 M4TESTS register
				184	* R13 (work)
				185	* R14 Subroutine call
				186	* R15 Secondary Subroutine call or work
				187	*
				188	*****
00000200		00000200		190	USING BEGIN,R8 FIRST Base Register
00000200		00001200		191	USING BEGIN+4096,R9 SECOND Base Register
00000200		00002200		192	USING BEGIN+8192,R10 THIRD Base Register
00000200	0580			194	BEGIN BALR R8,0 Inititalize FIRST base register
00000202	0680			195	BCTR R8,0 Inititalize FIRST base register
00000204	0680			196	BCTR R8,0 Inititalize FIRST base register
00000206	4190 8800		00000800	198	LA R9,2048(,R8) Inititalize SECOND base register
0000020A	4190 9800		00000800	199	LA R9,2048(,R9) Inititalize SECOND base register
				200	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT
				368 *****
				369 * Normal completion or Abnormal termination PSWs
				370 *****
00000458	00020001 80000000			372 EOJPSW DC 0D'0',X'0002000180000000',AD(0)
00000468	B2B2 8258		00000458	374 EOJ LPSWE EOJPSW Normal completion
00000470	00020001 80000000			376 FAILPSW DC 0D'0',X'0002000180000000',AD(X'BAD')
00000480	B2B2 8270		00000470	378 FAILTEST LPSWE FAILPSW Abnormal termination
				380 *****
				381 * Working Storage
				382 *****
00000484	00000000			384 CTLR0 DS F CR0
00000488	00000000			385 DS F
0000048C				387 LTORG , Literals pool
0000048C	00000008			388 =F'8'
00000490	00001BB8			389 =A(M4TESTS)
00000494	00000001			390 =F'1'
00000498	0000			391 =H'0'
0000049A	005F			392 =AL2(L'MSGMSG)
				393
				394 * some constants
				395
	00000400	00000001		396 K EQU 1024 One KB
	00001000	00000001		397 PAGE EQU (4*K) Size of one page
	00010000	00000001		398 K64 EQU (64*K) 64 KB
	00100000	00000001		399 MB EQU (K*K) 1 MB
				400
	AABBCCDD	00000001		401 REG2PATT EQU X'AABBCCDD' Polluted Register pattern
	000000DD	00000001		402 REG2LOW EQU X'DD' (last byte above)

LOC	OBJECT	CODE	ADDR1	ADDR2	STMT
					442 *****
					443 *M4TEST DSECT
					444 *****
					446 M4TEST DSECT ,
00000000	00000000				447 TSUB DC A(0)pointer to test
00000004	0000				448 TNUM DC H'00'Test Number
00000006	00				449 DC X'00'
					450
00000007	40404040	40404040			451 OPNAME DC CL8' 'E7 name
00000010	00000000				452 R2ADDR DC A(0)address of R2 source
00000014	00000000				453 R3ADDR DC A(0)address of R3 source
00000018	00000000				454 RELEN DC A(0)RESULT LENGTH
0000001C	00000000				455 READDR DC A(0)result (expected) address
00000020	00000000	00000000			456 DS FDgap
00000028	00000000	00000000			457 R10UTPUT DS Dr1 Output
00000030	00000000	00000000			458 DS FDgap
					459
					460 *test routine will be here (from RRF-a macro)
					461 *
					462 *followed by
					463 *EXPECTED RESULT
000010A8			00000000	00001C1F	465 MIE4TST CSECT ,
					466 DS 0F
					468 *****
					469 *Macros to help build test tables
					470 *****
					472 *
					473 *macro to generate individual test
					474 *
					475 MACRO
					476 RRF_A &INST
					477 .*&INST - RRF_A instruction under test
					478
					479
					480 GBLA &TNUM
					481 &TNUM SETA &TNUM+1
					482
					483 DS 0FD
					484 USING *,R5base for test data and test routine
					485
					486 T&TNUM DC A(X&TNUM)address of test routine
					487 DC H'&TNUM'test number
					488 DC X'00'
					489 DC CL8'&INST' instruction name
					490 DC A(RE&TNUM+8)address of R2 source
					491 DC A(RE&TNUM+16)address of R3 source
					492 DC A(8)result length

LOC	OBJECT CODE	ADDR1	ADDR2	STMT	
				543 *****	
				544 * MIE4 RRF_A tests	
				545 *****	
000010A8	00000000 00000000			546 PRINT DATA	
				547 DS FD	
				548 *	
				549 * B96C BEXTG - Bit Extract	
				550 * B96D BDEPG - Bit Deposit	
				551 *	
				552 * RRF_A instruction	
				553 * followed by	
				554 * 8 byte expected result (R1)	
				555 * 8 byte R2 source	
				556 * 8 byte R3 source	
				557 * -----	
				558 * BEXTG - Bit Extract	
				559 * -----	
				560 *	
				561	
000010B0				562 RRF_A BEXTG	
000010B0		000010B0		563+ DS 0FD	
000010B0	000010E8			564+ USING *,R5	base for test data and test routine
000010B4	0001			565+T1 DC A(X1)	address of test routine
000010B6	00			566+ DC H'1'	test number
000010B7	C2C5E7E3 C7404040			567+ DC X'00'	
000010C0	00001120			568+ DC CL8'BEXTG'	instruction name
000010C4	00001128			569+ DC A(RE1+8)	address of R2 source
000010C8	00000008			570+ DC A(RE1+16)	address of R3 source
000010CC	00001118			571+ DC A(8)	result length
000010D0	00000000 00000000			572+REA1 DC A(RE1)	result address
000010D8	00000000 00000000			573+ DS FD	gap
000010E0	00000000 00000000			574+R101 DS D	R1 output
				575+ DS FD	gap
				576+*	
000010E8				577+X1 DS 0F	
000010E8	E310 8E88 0004	00001088		578+ LG R1,R1FUDGE	
000010EE	B982 0022			579+ XGR R2,R2	
000010F2	5820 5010	00000010		580+ L R2,R2ADDR	load r2 source address
000010F6	E322 0000 0004	00000000		581+ LG R2,0(R2)	load source
000010FC	B982 0033			582+ XGR R3,R3	
00001100	5830 5014	00000014		583+ L R3,R3ADDR	load r3 source address
00001104	E333 0000 0004	00000000		584+ LG R3,0(R3)	load source
0000110A				587+ DS 0H	
0000110A	B96C3012			588+ DC XL4'B96C3012'	BEXTG r1,r2,R3
0000110E	E310 5028 0024	000010D8		590+ STG R1,R101	save R1 output
00001114	07FB			591+ BR R11	return
00001118				592+RE1 DC 0F	xl16 expected result
00001118				593+ DROP R5	
00001118	00000000 00000000			594 DC XL8'0000 0000 0000 0000'	result
00001120	AAAAAAAA AAAAAAAAAA			595 DC XL8'AAAA AAAA AAAA AAAA'	R2
00001128	00000000 00000000			596 DC XL8'0000 0000 0000 0000'	R3
				597	
				598 RRF_A BEXTG	
00001130				599+ DS 0FD	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
00001130		00001130		600+	USING *,R5	base for test data and test routine
00001130	00001168			601+T2	DC A(X2)	address of test routine
00001134	0002			602+	DC H'2'	test number
00001136	00			603+	DC X'00'	
00001137	C2C5E7E3 C7404040			604+	DC CL8'BEXTG'	instruction name
00001140	000011A0			605+	DC A(RE2+8)	address of R2 source
00001144	000011A8			606+	DC A(RE2+16)	address of R3 source
00001148	00000008			607+	DC A(8)	result length
0000114C	00001198			608+REA2	DC A(RE2)	result address
00001150	00000000 00000000			609+	DS FD	gap
00001158	00000000 00000000			610+R102	DS D	R1 output
00001160	00000000 00000000			611+	DS FD	gap
				612+*		
00001168				613+X2	DS 0F	
00001168	E310 8E88 0004		00001088	614+	LG R1,R1FUDGE	
0000116E	B982 0022			615+	XGR R2,R2	
00001172	5820 5010		00000010	616+	L R2,R2ADDR	load r2 source address
00001176	E322 0000 0004		00000000	617+	LG R2,0(R2)	load source
0000117C	B982 0033			618+	XGR R3,R3	
00001180	5830 5014		00000014	619+	L R3,R3ADDR	load r3 source address
00001184	E333 0000 0004		00000000	620+	LG R3,0(R3)	load source
0000118A				623+	DS 0H	
0000118A	B96C3012			624+	DC XL4'B96C3012'	BEXTG r1,r2,R3
0000118E	E310 5028 0024		00001158	626+	STG R1,R102	save R1 output
00001194	07FB			627+	BR R11	return
00001198				628+RE2	DC 0F	xl16 expected result
00001198				629+	DROP R5	
00001198	00000000 00000000			630	DC XL8'0000 0000 0000 0000'	result
000011A0	00000000 00000000			631	DC XL8'0000 0000 0000 0000'	R2
000011A8	AAAAAAAA AAAAAAAAAA			632	DC XL8'AAAA AAAA AAAA AAAA'	R3
				633		
				634	RRF_A BEXTG	
000011B0				635+	DS 0FD	
000011B0		000011B0		636+	USING *,R5	base for test data and test routine
000011B0	000011E8			637+T3	DC A(X3)	address of test routine
000011B4	0003			638+	DC H'3'	test number
000011B6	00			639+	DC X'00'	
000011B7	C2C5E7E3 C7404040			640+	DC CL8'BEXTG'	instruction name
000011C0	00001220			641+	DC A(RE3+8)	address of R2 source
000011C4	00001228			642+	DC A(RE3+16)	address of R3 source
000011C8	00000008			643+	DC A(8)	result length
000011CC	00001218			644+REA3	DC A(RE3)	result address
000011D0	00000000 00000000			645+	DS FD	gap
000011D8	00000000 00000000			646+R103	DS D	R1 output
000011E0	00000000 00000000			647+	DS FD	gap
				648+*		
000011E8				649+X3	DS 0F	
000011E8	E310 8E88 0004		00001088	650+	LG R1,R1FUDGE	
000011EE	B982 0022			651+	XGR R2,R2	
000011F2	5820 5010		00000010	652+	L R2,R2ADDR	load r2 source address
000011F6	E322 0000 0004		00000000	653+	LG R2,0(R2)	load source
000011FC	B982 0033			654+	XGR R3,R3	
00001200	5830 5014		00000014	655+	L R3,R3ADDR	load r3 source address
00001204	E333 0000 0004		00000000	656+	LG R3,0(R3)	load source

LOC	OBJECT CODE	ADDR1	ADDR2	STMT					
0000120A				659+	DS	0H			
0000120A	B96C3012			660+	DC	XL4'B96C3012'		BEXTG r1,r2,R3	
0000120E	E310 5028 0024		000011D8	662+	STG	R1,R103		save R1 output	
00001214	07FB			663+	BR	R11		return	
00001218				664+RE3	DC	0F		xl16 expected result	
00001218				665+	DROP	R5			
00001218	B8000000 00000000			666	DC	XL8'B800 0000 0000 0000'		result	
00001220	00089000 000AB000			667	DC	XL8'0008 9000 000A B000'		R2	
00001228	00080001 000A8000			668	DC	XL8'0008 0001 000A 8000'		R3	
				669					
00001230				670	RRF_A	BEXTG			
00001230		00001230		671+	DS	0FD			
00001230	00001268			672+	USING	*,R5		base for test data and test routine	
00001230	0004			673+T4	DC	A(X4)		address of test routine	
00001234	00			674+	DC	H'4'		test number	
00001236	00			675+	DC	X'00'			
00001237	C2C5E7E3 C7404040			676+	DC	CL8'BEXTG'		instruction name	
00001240	000012A0			677+	DC	A(RE4+8)		address of R2 source	
00001244	000012A8			678+	DC	A(RE4+16)		address of R3 source	
00001248	00000008			679+	DC	A(8)		result length	
0000124C	00001298			680+REA4	DC	A(RE4)		result address	
00001250	00000000 00000000			681+	DS	FD		gap	
00001258	00000000 00000000			682+R104	DS	D		R1 output	
00001260	00000000 00000000			683+	DS	FD		gap	
				684+*					
00001268				685+X4	DS	0F			
00001268	E310 8E88 0004		00001088	686+	LG	R1,R1FUDGE			
0000126E	B982 0022			687+	XGR	R2,R2			
00001272	5820 5010		00000010	688+	L	R2,R2ADDR		load r2 source address	
00001276	E322 0000 0004		00000000	689+	LG	R2,0(R2)		load source	
0000127C	B982 0033			690+	XGR	R3,R3			
00001280	5830 5014		00000014	691+	L	R3,R3ADDR		load r3 source address	
00001284	E333 0000 0004		00000000	692+	LG	R3,0(R3)		load source	
0000128A				695+	DS	0H			
0000128A	B96C3012			696+	DC	XL4'B96C3012'		BEXTG r1,r2,R3	
0000128E	E310 5028 0024		00001258	698+	STG	R1,R104		save R1 output	
00001294	07FB			699+	BR	R11		return	
00001298				700+RE4	DC	0F		xl16 expected result	
00001298				701+	DROP	R5			
00001298	9A000000 00000000			702	DC	XL8'9A00 0000 0000 0000'		result	
000012A0	00089000 000AB000			703	DC	XL8'0008 9000 000A B000'		R2	
000012A8	0008000C 000A6006			704	DC	XL8'0008 000C 000A 6006'		R3	
				705					
				706	RRF_A	BEXTG			
000012B0				707+	DS	0FD			
000012B0		000012B0		708+	USING	*,R5		base for test data and test routine	
000012B0	000012E8			709+T5	DC	A(X5)		address of test routine	
000012B4	0005			710+	DC	H'5'		test number	
000012B6	00			711+	DC	X'00'			
000012B7	C2C5E7E3 C7404040			712+	DC	CL8'BEXTG'		instruction name	
000012C0	00001320			713+	DC	A(RE5+8)		address of R2 source	
000012C4	00001328			714+	DC	A(RE5+16)		address of R3 source	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
000012C8	00000008			715+	DC	A(8)	result length
000012CC	00001318			716+REA5	DC	A(RE5)	result address
000012D0	00000000 00000000			717+	DS	FD	gap
000012D8	00000000 00000000			718+R105	DS	D	R1 output
000012E0	00000000 00000000			719+	DS	FD	gap
				720+*			
000012E8				721+X5	DS	0F	
000012E8	E310 8E88 0004		00001088	722+	LG	R1,R1FUDGE	
000012EE	B982 0022			723+	XGR	R2,R2	
000012F2	5820 5010		00000010	724+	L	R2,R2ADDR	load r2 source address
000012F6	E322 0000 0004		00000000	725+	LG	R2,0(R2)	load source
000012FC	B982 0033			726+	XGR	R3,R3	
00001300	5830 5014		00000014	727+	L	R3,R3ADDR	load r3 source address
00001304	E333 0000 0004		00000000	728+	LG	R3,0(R3)	load source
0000130A				731+	DS	0H	
0000130A	B96C3012			732+	DC	XL4'B96C3012'	BEXTG r1,r2,R3
0000130E	E310 5028 0024		000012D8	734+	STG	R1,R105	save R1 output
00001314	07FB			735+	BR	R11	return
00001318				736+RE5	DC	0F	xl16 expected result
00001318				737+	DROP	R5	
00001318	9B000000 00000000			738	DC	XL8'9B00 0000 0000 0000'	result
00001320	00089000 000AB004			739	DC	XL8'0008 9000 000A B004'	R2
00001328	0008000C 000A6006			740	DC	XL8'0008 000C 000A 6006'	R3
				741			
				742	RRF_A	BEXTG	
00001330				743+	DS	0FD	
00001330		00001330		744+	USING	*,R5	base for test data and test routine
00001330	00001368			745+T6	DC	A(X6)	address of test routine
00001334	0006			746+	DC	H'6'	test number
00001336	00			747+	DC	X'00'	
00001337	C2C5E7E3 C7404040			748+	DC	CL8'BEXTG'	instruction name
00001340	000013A0			749+	DC	A(RE6+8)	address of R2 source
00001344	000013A8			750+	DC	A(RE6+16)	address of R3 source
00001348	00000008			751+	DC	A(8)	result length
0000134C	00001398			752+REA6	DC	A(RE6)	result address
00001350	00000000 00000000			753+	DS	FD	gap
00001358	00000000 00000000			754+R106	DS	D	R1 output
00001360	00000000 00000000			755+	DS	FD	gap
				756+*			
00001368				757+X6	DS	0F	
00001368	E310 8E88 0004		00001088	758+	LG	R1,R1FUDGE	
0000136E	B982 0022			759+	XGR	R2,R2	
00001372	5820 5010		00000010	760+	L	R2,R2ADDR	load r2 source address
00001376	E322 0000 0004		00000000	761+	LG	R2,0(R2)	load source
0000137C	B982 0033			762+	XGR	R3,R3	
00001380	5830 5014		00000014	763+	L	R3,R3ADDR	load r3 source address
00001384	E333 0000 0004		00000000	764+	LG	R3,0(R3)	load source
0000138A				767+	DS	0H	
0000138A	B96C3012			768+	DC	XL4'B96C3012'	BEXTG r1,r2,R3
0000138E	E310 5028 0024		00001358	770+	STG	R1,R106	save R1 output
00001394	07FB			771+	BR	R11	return
00001398				772+RE6	DC	0F	xl16 expected result

LOC	OBJECT CODE	ADDR1	ADDR2	STMT						
00001398				773+	DROP	R5				
00001398	9B800000 00000000			774	DC	XL8'9B80 0000 0000 0000'	result			
000013A0	00089000 000AB006			775	DC	XL8'0008 9000 000A B006'	R2			
000013A8	0008000C 000A6006			776	DC	XL8'0008 000C 000A 6006'	R3			
				777						
				778	RRF_A	BEXTG				
000013B0				779+	DS	0FD				
000013B0		000013B0		780+	USING	*,R5	base for test data and test routine			
000013B0	000013E8			781+T7	DC	A(X7)	address of test routine			
000013B4	0007			782+	DC	H'7'	test number			
000013B6	00			783+	DC	X'00'				
000013B7	C2C5E7E3 C7404040			784+	DC	CL8'BEXTG'	instruction name			
000013C0	00001420			785+	DC	A(RE7+8)	address of R2 source			
000013C4	00001428			786+	DC	A(RE7+16)	address of R3 source			
000013C8	00000008			787+	DC	A(8)	result length			
000013CC	00001418			788+REA7	DC	A(RE7)	result address			
000013D0	00000000 00000000			789+	DS	FD	gap			
000013D8	00000000 00000000			790+R107	DS	D	R1 output			
000013E0	00000000 00000000			791+	DS	FD	gap			
				792+*						
000013E8				793+X7	DS	0F				
000013E8	E310 8E88 0004		00001088	794+	LG	R1,R1FUDGE				
000013EE	B982 0022			795+	XGR	R2,R2				
000013F2	5820 5010		00000010	796+	L	R2,R2ADDR	load r2 source address			
000013F6	E322 0000 0004		00000000	797+	LG	R2,0(R2)	load source			
000013FC	B982 0033			798+	XGR	R3,R3				
00001400	5830 5014		00000014	799+	L	R3,R3ADDR	load r3 source address			
00001404	E333 0000 0004		00000000	800+	LG	R3,0(R3)	load source			
0000140A				803+	DS	0H				
0000140A	B96C3012			804+	DC	XL4'B96C3012'	BEXTG r1,r2,R3			
0000140E	E310 5028 0024		000013D8	806+	STG	R1,R107	save R1 output			
00001414	07FB			807+	BR	R11	return			
00001418				808+RE7	DC	0F	xl16 expected result			
00001418				809+	DROP	R5				
00001418	89AB0000 00000000			810	DC	XL8'89AB 0000 0000 0000'	result			
00001420	00089000 000AB000			811	DC	XL8'0008 9000 000A B000'	R2			
00001428	000FF000 000FF000			812	DC	XL8'000F F000 000F F000'	R3			
				813						
				814	RRF_A	BEXTG				
00001430				815+	DS	0FD				
00001430		00001430		816+	USING	*,R5	base for test data and test routine			
00001430	00001468			817+T8	DC	A(X8)	address of test routine			
00001434	0008			818+	DC	H'8'	test number			
00001436	00			819+	DC	X'00'				
00001437	C2C5E7E3 C7404040			820+	DC	CL8'BEXTG'	instruction name			
00001440	000014A0			821+	DC	A(RE8+8)	address of R2 source			
00001444	000014A8			822+	DC	A(RE8+16)	address of R3 source			
00001448	00000008			823+	DC	A(8)	result length			
0000144C	00001498			824+REA8	DC	A(RE8)	result address			
00001450	00000000 00000000			825+	DS	FD	gap			
00001458	00000000 00000000			826+R108	DS	D	R1 output			
00001460	00000000 00000000			827+	DS	FD	gap			
				828+*						
00001468				829+X8	DS	0F				

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00001468	E310 8E88 0004		00001088	830+	LG	R1,R1FUDGE	
0000146E	B982 0022			831+	XGR	R2,R2	
00001472	5820 5010		00000010	832+	L	R2,R2ADDR	load r2 source address
00001476	E322 0000 0004		00000000	833+	LG	R2,0(R2)	load source
0000147C	B982 0033			834+	XGR	R3,R3	
00001480	5830 5014		00000014	835+	L	R3,R3ADDR	load r3 source address
00001484	E333 0000 0004		00000000	836+	LG	R3,0(R3)	load source
0000148A				839+	DS	0H	
0000148A	B96C3012			840+	DC	XL4'B96C3012'	BEXTG r1,r2,R3
0000148E	E310 5028 0024		00001458	842+	STG	R1,R108	save R1 output
00001494	07FB			843+	BR	R11	return
00001498				844+RE8	DC	0F	xl16 expected result
00001498				845+	DROP	R5	
00001498	FFFF0000 00000000			846	DC	XL8'FFFF 0000 0000 0000'	result
000014A0	000FF000 000FF000			847	DC	XL8'000F F000 000F F000'	R2
000014A8	000FF000 000FF000			848	DC	XL8'000F F000 000F F000'	R3
				849			
				850	RRF_A	BEXTG	
000014B0				851+	DS	0FD	
000014B0		000014B0		852+	USING	*,R5	base for test data and test routine
000014B0	000014E8			853+T9	DC	A(X9)	address of test routine
000014B4	0009			854+	DC	H'9'	test number
000014B6	00			855+	DC	X'00'	
000014B7	C2C5E7E3 C7404040			856+	DC	CL8'BEXTG'	instruction name
000014C0	00001520			857+	DC	A(RE9+8)	address of R2 source
000014C4	00001528			858+	DC	A(RE9+16)	address of R3 source
000014C8	00000008			859+	DC	A(8)	result length
000014CC	00001518			860+REA9	DC	A(RE9)	result address
000014D0	00000000 00000000			861+	DS	FD	gap
000014D8	00000000 00000000			862+R109	DS	D	R1 output
000014E0	00000000 00000000			863+	DS	FD	gap
				864+*			
000014E8				865+X9	DS	0F	
000014E8	E310 8E88 0004		00001088	866+	LG	R1,R1FUDGE	
000014EE	B982 0022			867+	XGR	R2,R2	
000014F2	5820 5010		00000010	868+	L	R2,R2ADDR	load r2 source address
000014F6	E322 0000 0004		00000000	869+	LG	R2,0(R2)	load source
000014FC	B982 0033			870+	XGR	R3,R3	
00001500	5830 5014		00000014	871+	L	R3,R3ADDR	load r3 source address
00001504	E333 0000 0004		00000000	872+	LG	R3,0(R3)	load source
0000150A				875+	DS	0H	
0000150A	B96C3012			876+	DC	XL4'B96C3012'	BEXTG r1,r2,R3
0000150E	E310 5028 0024		000014D8	878+	STG	R1,R109	save R1 output
00001514	07FB			879+	BR	R11	return
00001518				880+RE9	DC	0F	xl16 expected result
00001518				881+	DROP	R5	
00001518	00000000 00000000			882	DC	XL8'0000 0000 0000 0000'	result
00001520	00000000 00000000			883	DC	XL8'0000 0000 0000 0000'	R2
00001528	FFFFFFFF FFFFFFFF			884	DC	XL8'FFFF FFFF FFFF FFFF'	R3
				885			
				886	RRF_A	BEXTG	
00001530				887+	DS	0FD	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
00001530		00001530		888+	USING *,R5	base for test data and test routine
00001530	00001568			889+T10	DC A(X10)	address of test routine
00001534	000A			890+	DC H'10'	test number
00001536	00			891+	DC X'00'	
00001537	C2C5E7E3 C7404040			892+	DC CL8'BEXTG'	instruction name
00001540	000015A0			893+	DC A(RE10+8)	address of R2 source
00001544	000015A8			894+	DC A(RE10+16)	address of R3 source
00001548	00000008			895+	DC A(8)	result length
0000154C	00001598			896+REA10	DC A(RE10)	result address
00001550	00000000 00000000			897+	DS FD	gap
00001558	00000000 00000000			898+R1010	DS D	R1 output
00001560	00000000 00000000			899+	DS FD	gap
				900+*		
00001568				901+X10	DS 0F	
00001568	E310 8E88 0004		00001088	902+	LG R1,R1FUDGE	
0000156E	B982 0022			903+	XGR R2,R2	
00001572	5820 5010		00000010	904+	L R2,R2ADDR	load r2 source address
00001576	E322 0000 0004		00000000	905+	LG R2,0(R2)	load source
0000157C	B982 0033			906+	XGR R3,R3	
00001580	5830 5014		00000014	907+	L R3,R3ADDR	load r3 source address
00001584	E333 0000 0004		00000000	908+	LG R3,0(R3)	load source
0000158A				911+	DS 0H	
0000158A	B96C3012			912+	DC XL4'B96C3012'	BEXTG r1,r2,R3
0000158E	E310 5028 0024		00001558	914+	STG R1,R1010	save R1 output
00001594	07FB			915+	BR R11	return
00001598				916+RE10	DC 0F	xl16 expected result
00001598				917+	DROP R5	
00001598	10001000 10000001			918	DC XL8'1000 1000 1000 0001'	result
000015A0	10001000 10000001			919	DC XL8'1000 1000 1000 0001'	R2
000015A8	FFFFFFFF FFFFFFFF			920	DC XL8'FFFF FFFF FFFF FFFF'	R3
				921		
				922	RRF_A BEXTG	
000015B0				923+	DS 0FD	
000015B0		000015B0		924+	USING *,R5	base for test data and test routine
000015B0	000015E8			925+T11	DC A(X11)	address of test routine
000015B4	000B			926+	DC H'11'	test number
000015B6	00			927+	DC X'00'	
000015B7	C2C5E7E3 C7404040			928+	DC CL8'BEXTG'	instruction name
000015C0	00001620			929+	DC A(RE11+8)	address of R2 source
000015C4	00001628			930+	DC A(RE11+16)	address of R3 source
000015C8	00000008			931+	DC A(8)	result length
000015CC	00001618			932+REA11	DC A(RE11)	result address
000015D0	00000000 00000000			933+	DS FD	gap
000015D8	00000000 00000000			934+R1011	DS D	R1 output
000015E0	00000000 00000000			935+	DS FD	gap
				936+*		
000015E8				937+X11	DS 0F	
000015E8	E310 8E88 0004		00001088	938+	LG R1,R1FUDGE	
000015EE	B982 0022			939+	XGR R2,R2	
000015F2	5820 5010		00000010	940+	L R2,R2ADDR	load r2 source address
000015F6	E322 0000 0004		00000000	941+	LG R2,0(R2)	load source
000015FC	B982 0033			942+	XGR R3,R3	
00001600	5830 5014		00000014	943+	L R3,R3ADDR	load r3 source address
00001604	E333 0000 0004		00000000	944+	LG R3,0(R3)	load source

LOC	OBJECT CODE	ADDR1	ADDR2	STMT					
0000160A				947+	DS	0H			
0000160A	B96C3012			948+	DC	XL4'B96C3012'		BEXTG r1,r2,R3	
0000160E	E310 5028 0024		000015D8	950+	STG	R1,R1011		save R1 output	
00001614	07FB			951+	BR	R11		return	
00001618				952+RE11	DC	0F		xl16 expected result	
00001618				953+	DROP	R5			
00001618	AAAAAAAA AAAAAAAAAA			954	DC	XL8'AAAA AAAA AAAA AAAA'		result	
00001620	AAAAAAAA AAAAAAAAAA			955	DC	XL8'AAAA AAAA AAAA AAAA'		R2	
00001628	FFFFFFFF FFFFFFFF			956	DC	XL8'FFFF FFFF FFFF FFFF'		R3	
				957					
				958 *					
				959 *					
				960 *					
				961 *					
				962	RRF_A	BDEPG			
00001630				963+	DS	0FD			
00001630		00001630		964+	USING	*,R5		base for test data and test routine	
00001630	00001668			965+T12	DC	A(X12)		address of test routine	
00001634	000C			966+	DC	H'12'		test number	
00001636	00			967+	DC	X'00'			
00001637	C2C4C5D7 C7404040			968+	DC	CL8'BDEPG'		instruction name	
00001640	000016A0			969+	DC	A(RE12+8)		address of R2 source	
00001644	000016A8			970+	DC	A(RE12+16)		address of R3 source	
00001648	00000008			971+	DC	A(8)		result length	
0000164C	00001698			972+REA12	DC	A(RE12)		result address	
00001650	00000000 00000000			973+	DS	FD		gap	
00001658	00000000 00000000			974+R1012	DS	D		R1 output	
00001660	00000000 00000000			975+	DS	FD		gap	
				976+*					
00001668				977+X12	DS	0F			
00001668	E310 8E88 0004		00001088	978+	LG	R1,R1FUDGE			
0000166E	B982 0022			979+	XGR	R2,R2			
00001672	5820 5010		00000010	980+	L	R2,R2ADDR		load r2 source address	
00001676	E322 0000 0004		00000000	981+	LG	R2,0(R2)		load source	
0000167C	B982 0033			982+	XGR	R3,R3			
00001680	5830 5014		00000014	983+	L	R3,R3ADDR		load r3 source address	
00001684	E333 0000 0004		00000000	984+	LG	R3,0(R3)		load source	
0000168A				987+	DS	0H			
0000168A	B96D3012			988+	DC	XL4'B96D3012'		BDEPG r1,r2,R3	
0000168E	E310 5028 0024		00001658	990+	STG	R1,R1012		save R1 output	
00001694	07FB			991+	BR	R11		return	
00001698				992+RE12	DC	0F		xl16 expected result	
00001698				993+	DROP	R5			
00001698	00000000 00000000			994	DC	XL8'0000 0000 0000 0000'		result	
000016A0	AAAAAAAA AAAAAAAAAA			995	DC	XL8'AAAA AAAA AAAA AAAA'		R2	
000016A8	00000000 00000000			996	DC	XL8'0000 0000 0000 0000'		R3	
				997					
				998	RRF_A	BDEPG			
000016B0				999+	DS	0FD			
000016B0		000016B0		1000+	USING	*,R5		base for test data and test routine	
000016B0	000016E8			1001+T13	DC	A(X13)		address of test routine	
000016B4	000D			1002+	DC	H'13'		test number	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
000016B6	00			1003+	DC	X'00'	
000016B7	C2C4C5D7 C7404040			1004+	DC	CL8'BDEPG'	instruction name
000016C0	00001720			1005+	DC	A(RE13+8)	address of R2 source
000016C4	00001728			1006+	DC	A(RE13+16)	address of R3 source
000016C8	00000008			1007+	DC	A(8)	result length
000016CC	00001718			1008+REA13	DC	A(RE13)	result address
000016D0	00000000 00000000			1009+	DS	FD	gap
000016D8	00000000 00000000			1010+R1013	DS	D	R1 output
000016E0	00000000 00000000			1011+	DS	FD	gap
				1012+*			
000016E8				1013+X13	DS	0F	
000016E8	E310 8E88 0004		00001088	1014+	LG	R1,R1FUDGE	
000016EE	B982 0022			1015+	XGR	R2,R2	
000016F2	5820 5010		00000010	1016+	L	R2,R2ADDR	load r2 source address
000016F6	E322 0000 0004		00000000	1017+	LG	R2,0(R2)	load source
000016FC	B982 0033			1018+	XGR	R3,R3	
00001700	5830 5014		00000014	1019+	L	R3,R3ADDR	load r3 source address
00001704	E333 0000 0004		00000000	1020+	LG	R3,0(R3)	load source
0000170A				1023+	DS	0H	
0000170A	B96D3012			1024+	DC	XL4'B96D3012'	BDEPG r1,r2,R3
0000170E	E310 5028 0024		000016D8	1026+	STG	R1,R1013	save R1 output
00001714	07FB			1027+	BR	R11	return
00001718				1028+RE13	DC	0F	xl16 expected result
00001718				1029+	DROP	R5	
00001718	00000000 00000000			1030	DC	XL8'0000 0000 0000 0000'	result
00001720	00000000 00000000			1031	DC	XL8'0000 0000 0000 0000'	R2
00001728	AAAAAAAA AAAAAAAAAA			1032	DC	XL8'AAAA AAAA AAAA AAAA'	R3
				1033			
00001730				1034	RRF_A	BDEPG	
00001730		00001730		1035+	DS	0FD	
00001730	00001768			1036+	USING	*,R5	base for test data and test routine
00001734	000E			1037+T14	DC	A(X14)	address of test routine
00001736	00			1038+	DC	H'14'	test number
00001736	00			1039+	DC	X'00'	
00001737	C2C4C5D7 C7404040			1040+	DC	CL8'BDEPG'	instruction name
00001740	000017A0			1041+	DC	A(RE14+8)	address of R2 source
00001744	000017A8			1042+	DC	A(RE14+16)	address of R3 source
00001748	00000008			1043+	DC	A(8)	result length
0000174C	00001798			1044+REA14	DC	A(RE14)	result address
00001750	00000000 00000000			1045+	DS	FD	gap
00001758	00000000 00000000			1046+R1014	DS	D	R1 output
00001760	00000000 00000000			1047+	DS	FD	gap
				1048+*			
00001768				1049+X14	DS	0F	
00001768	E310 8E88 0004		00001088	1050+	LG	R1,R1FUDGE	
0000176E	B982 0022			1051+	XGR	R2,R2	
00001772	5820 5010		00000010	1052+	L	R2,R2ADDR	load r2 source address
00001776	E322 0000 0004		00000000	1053+	LG	R2,0(R2)	load source
0000177C	B982 0033			1054+	XGR	R3,R3	
00001780	5830 5014		00000014	1055+	L	R3,R3ADDR	load r3 source address
00001784	E333 0000 0004		00000000	1056+	LG	R3,0(R3)	load source
0000178A				1059+	DS	0H	
0000178A	B96D3012			1060+	DC	XL4'B96D3012'	BDEPG r1,r2,R3

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
0000178E	E310 5028 0024		00001758	1062+	STG	R1,R1014	save R1 output
00001794	07FB			1063+	BR	R11	return
00001798				1064+RE14	DC	0F	xl16 expected result
00001798				1065+	DROP	R5	
00001798	00080000 000A8000			1066	DC	XL8'0008 0000 000A 8000'	result
000017A0	B8000000 00000000			1067	DC	XL8'B800 0000 0000 0000'	R2
000017A8	00080001 000A8000			1068	DC	XL8'0008 0001 000A 8000'	R3
				1069			
				1070	RRF_A	BDEPG	
000017B0				1071+	DS	0FD	
000017B0		000017B0		1072+	USING	*,R5	base for test data and test routine
000017B0	000017E8			1073+T15	DC	A(X15)	address of test routine
000017B4	000F			1074+	DC	H'15'	test number
000017B6	00			1075+	DC	X'00'	
000017B7	C2C4C5D7 C7404040			1076+	DC	CL8'BDEPG'	instruction name
000017C0	00001820			1077+	DC	A(RE15+8)	address of R2 source
000017C4	00001828			1078+	DC	A(RE15+16)	address of R3 source
000017C8	00000008			1079+	DC	A(8)	result length
000017CC	00001818			1080+REA15	DC	A(RE15)	result address
000017D0	00000000 00000000			1081+	DS	FD	gap
000017D8	00000000 00000000			1082+R1015	DS	D	R1 output
000017E0	00000000 00000000			1083+	DS	FD	gap
				1084+*			
000017E8				1085+X15	DS	0F	
000017E8	E310 8E88 0004		00001088	1086+	LG	R1,R1FUDGE	
000017EE	B982 0022			1087+	XGR	R2,R2	
000017F2	5820 5010		00000010	1088+	L	R2,R2ADDR	load r2 source address
000017F6	E322 0000 0004		00000000	1089+	LG	R2,0(R2)	load source
000017FC	B982 0033			1090+	XGR	R3,R3	
00001800	5830 5014		00000014	1091+	L	R3,R3ADDR	load r3 source address
00001804	E333 0000 0004		00000000	1092+	LG	R3,0(R3)	load source
0000180A				1095+	DS	0H	
0000180A	B96D3012			1096+	DC	XL4'B96D3012'	BDEPG r1,r2,R3
0000180E	E310 5028 0024		000017D8	1098+	STG	R1,R1015	save R1 output
00001814	07FB			1099+	BR	R11	return
00001818				1100+RE15	DC	0F	xl16 expected result
00001818				1101+	DROP	R5	
00001818	00080000 000A2000			1102	DC	XL8'0008 0000 000A 2000'	result
00001820	9A000000 00000000			1103	DC	XL8'9A00 0000 0000 0000'	R2
00001828	0008000C 000A6006			1104	DC	XL8'0008 000C 000A 6006'	R3
				1105			
				1106	RRF_A	BDEPG	
00001830				1107+	DS	0FD	
00001830		00001830		1108+	USING	*,R5	base for test data and test routine
00001830	00001868			1109+T16	DC	A(X16)	address of test routine
00001834	0010			1110+	DC	H'16'	test number
00001836	00			1111+	DC	X'00'	
00001837	C2C4C5D7 C7404040			1112+	DC	CL8'BDEPG'	instruction name
00001840	000018A0			1113+	DC	A(RE16+8)	address of R2 source
00001844	000018A8			1114+	DC	A(RE16+16)	address of R3 source
00001848	00000008			1115+	DC	A(8)	result length
0000184C	00001898			1116+REA16	DC	A(RE16)	result address
00001850	00000000 00000000			1117+	DS	FD	gap

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00001858	00000000 00000000			1118+R1016	DS	D	R1 output
00001860	00000000 00000000			1119+	DS	FD	gap
				1120+*			
00001868				1121+X16	DS	0F	
00001868	E310 8E88 0004		00001088	1122+	LG	R1,R1FUDGE	
0000186E	B982 0022			1123+	XGR	R2,R2	
00001872	5820 5010		00000010	1124+	L	R2,R2ADDR	load r2 source address
00001876	E322 0000 0004		00000000	1125+	LG	R2,0(R2)	load source
0000187C	B982 0033			1126+	XGR	R3,R3	
00001880	5830 5014		00000014	1127+	L	R3,R3ADDR	load r3 source address
00001884	E333 0000 0004		00000000	1128+	LG	R3,0(R3)	load source
0000188A				1131+	DS	0H	
0000188A	B96D3012			1132+	DC	XL4'B96D3012'	BDEPG r1,r2,R3
0000188E	E310 5028 0024		00001858	1134+	STG	R1,R1016	save R1 output
00001894	07FB			1135+	BR	R11	return
00001898				1136+RE16	DC	0F	xl16 expected result
00001898				1137+	DROP	R5	
00001898	00080000 000A2004			1138	DC	XL8'0008 0000 000A 2004'	result
000018A0	9B000000 00000000			1139	DC	XL8'9B00 0000 0000 0000'	R2
000018A8	0008000C 000A6006			1140	DC	XL8'0008 000C 000A 6006'	R3
				1141			
000018B0				1142	RRF_A	BDEPG	
000018B0		000018B0		1143+	DS	0FD	
000018B0	000018E8			1144+	USING	*,R5	base for test data and test routine
000018B4	0011			1145+T17	DC	A(X17)	address of test routine
000018B6	00			1146+	DC	H'17'	test number
000018B6	00			1147+	DC	X'00'	
000018B7	C2C4C5D7 C7404040			1148+	DC	CL8'BDEPG'	instruction name
000018C0	00001920			1149+	DC	A(RE17+8)	address of R2 source
000018C4	00001928			1150+	DC	A(RE17+16)	address of R3 source
000018C8	00000008			1151+	DC	A(8)	result length
000018CC	00001918			1152+REA17	DC	A(RE17)	result address
000018D0	00000000 00000000			1153+	DS	FD	gap
000018D8	00000000 00000000			1154+R1017	DS	D	R1 output
000018E0	00000000 00000000			1155+	DS	FD	gap
				1156+*			
000018E8				1157+X17	DS	0F	
000018E8	E310 8E88 0004		00001088	1158+	LG	R1,R1FUDGE	
000018EE	B982 0022			1159+	XGR	R2,R2	
000018F2	5820 5010		00000010	1160+	L	R2,R2ADDR	load r2 source address
000018F6	E322 0000 0004		00000000	1161+	LG	R2,0(R2)	load source
000018FC	B982 0033			1162+	XGR	R3,R3	
00001900	5830 5014		00000014	1163+	L	R3,R3ADDR	load r3 source address
00001904	E333 0000 0004		00000000	1164+	LG	R3,0(R3)	load source
0000190A				1167+	DS	0H	
0000190A	B96D3012			1168+	DC	XL4'B96D3012'	BDEPG r1,r2,R3
0000190E	E310 5028 0024		000018D8	1170+	STG	R1,R1017	save R1 output
00001914	07FB			1171+	BR	R11	return
00001918				1172+RE17	DC	0F	xl16 expected result
00001918				1173+	DROP	R5	
00001918	00080000 000A2006			1174	DC	XL8'0008 0000 000A 2006'	result
00001920	9B800000 00000000			1175	DC	XL8'9B80 0000 0000 0000'	R2

LOC	OBJECT CODE	ADDR1	ADDR2	STMT					
00001928	0008000C 000A6006			1176	DC	XL8'0008 000C 000A 6006'	R3		
				1177					
00001930				1178	RRF_A	BDEPG			
00001930		00001930		1179+	DS	0FD			
00001930	00001968			1180+	USING	*,R5		base for test data and test routine	
00001934	0012			1181+T18	DC	A(X18)		address of test routine	
00001936	00			1182+	DC	H'18'		test number	
00001937	C2C4C5D7 C7404040			1183+	DC	X'00'			
00001940	000019A0			1184+	DC	CL8'BDEPG'		instruction name	
00001944	000019A8			1185+	DC	A(RE18+8)		address of R2 source	
00001948	00000008			1186+	DC	A(RE18+16)		address of R3 source	
0000194C	00001998			1187+	DC	A(8)		result length	
00001950	00000000 00000000			1188+REA18	DC	A(RE18)		result address	
00001958	00000000 00000000			1189+	DS	FD		gap	
00001960	00000000 00000000			1190+R1018	DS	D		R1 output	
				1191+	DS	FD		gap	
				1192+*					
00001968				1193+X18	DS	0F			
00001968	E310 8E88 0004		00001088	1194+	LG	R1,R1FUDGE			
0000196E	B982 0022			1195+	XGR	R2,R2			
00001972	5820 5010		00000010	1196+	L	R2,R2ADDR		load r2 source address	
00001976	E322 0000 0004		00000000	1197+	LG	R2,0(R2)		load source	
0000197C	B982 0033			1198+	XGR	R3,R3			
00001980	5830 5014		00000014	1199+	L	R3,R3ADDR		load r3 source address	
00001984	E333 0000 0004		00000000	1200+	LG	R3,0(R3)		load source	
0000198A				1203+	DS	0H			
0000198A	B96D3012			1204+	DC	XL4'B96D3012'		BDEPG r1,r2,R3	
0000198E	E310 5028 0024		00001958	1206+	STG	R1,R1018		save R1 output	
00001994	07FB			1207+	BR	R11		return	
00001998				1208+RE18	DC	0F		xl16 expected result	
00001998				1209+	DROP	R5			
00001998	00089000 000AB000			1210	DC	XL8'0008 9000 000A B000'		result	
000019A0	89AB0000 00000000			1211	DC	XL8'89AB 0000 0000 0000'		R2	
000019A8	000FF000 000FF000			1212	DC	XL8'000F F000 000F F000'		R3	
				1213					
				1214	RRF_A	BDEPG			
000019B0				1215+	DS	0FD			
000019B0		000019B0		1216+	USING	*,R5		base for test data and test routine	
000019B0	000019E8			1217+T19	DC	A(X19)		address of test routine	
000019B4	0013			1218+	DC	H'19'		test number	
000019B6	00			1219+	DC	X'00'			
000019B7	C2C4C5D7 C7404040			1220+	DC	CL8'BDEPG'		instruction name	
000019C0	00001A20			1221+	DC	A(RE19+8)		address of R2 source	
000019C4	00001A28			1222+	DC	A(RE19+16)		address of R3 source	
000019C8	00000008			1223+	DC	A(8)		result length	
000019CC	00001A18			1224+REA19	DC	A(RE19)		result address	
000019D0	00000000 00000000			1225+	DS	FD		gap	
000019D8	00000000 00000000			1226+R1019	DS	D		R1 output	
000019E0	00000000 00000000			1227+	DS	FD		gap	
				1228+*					
000019E8				1229+X19	DS	0F			
000019E8	E310 8E88 0004		00001088	1230+	LG	R1,R1FUDGE			
000019EE	B982 0022			1231+	XGR	R2,R2			
000019F2	5820 5010		00000010	1232+	L	R2,R2ADDR		load r2 source address	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
000019F6	E322 0000 0004		00000000	1233+	LG	R2,0(R2)	load source
000019FC	B982 0033			1234+	XGR	R3,R3	
00001A00	5830 5014		00000014	1235+	L	R3,R3ADDR	load r3 source address
00001A04	E333 0000 0004		00000000	1236+	LG	R3,0(R3)	load source
00001A0A				1239+	DS	0H	
00001A0A	B96D3012			1240+	DC	XL4'B96D3012'	BDEPG r1,r2,R3
00001A0E	E310 5028 0024		000019D8	1242+	STG	R1,R1019	save R1 output
00001A14	07FB			1243+	BR	R11	return
00001A18				1244+RE19	DC	0F	xl16 expected result
00001A18				1245+	DROP	R5	
00001A18	000FF000 000FF000			1246	DC	XL8'000F F000 000F F000'	result
00001A20	FFFF0000 00000000			1247	DC	XL8'FFFF 0000 0000 0000'	R2
00001A28	000FF000 000FF000			1248	DC	XL8'000F F000 000F F000'	R3
				1249			
				1250	RRF_A	BDEPG	
00001A30				1251+	DS	0FD	
00001A30		00001A30		1252+	USING	*,R5	base for test data and test routine
00001A30	00001A68			1253+T20	DC	A(X20)	address of test routine
00001A34	0014			1254+	DC	H'20'	test number
00001A36	00			1255+	DC	X'00'	
00001A37	C2C4C5D7 C7404040			1256+	DC	CL8'BDEPG'	instruction name
00001A40	00001AA0			1257+	DC	A(RE20+8)	address of R2 source
00001A44	00001AA8			1258+	DC	A(RE20+16)	address of R3 source
00001A48	00000008			1259+	DC	A(8)	result length
00001A4C	00001A98			1260+REA20	DC	A(RE20)	result address
00001A50	00000000 00000000			1261+	DS	FD	gap
00001A58	00000000 00000000			1262+R1020	DS	D	R1 output
00001A60	00000000 00000000			1263+	DS	FD	gap
				1264+*			
00001A68				1265+X20	DS	0F	
00001A68	E310 8E88 0004		00001088	1266+	LG	R1,R1FUDGE	
00001A6E	B982 0022			1267+	XGR	R2,R2	
00001A72	5820 5010		00000010	1268+	L	R2,R2ADDR	load r2 source address
00001A76	E322 0000 0004		00000000	1269+	LG	R2,0(R2)	load source
00001A7C	B982 0033			1270+	XGR	R3,R3	
00001A80	5830 5014		00000014	1271+	L	R3,R3ADDR	load r3 source address
00001A84	E333 0000 0004		00000000	1272+	LG	R3,0(R3)	load source
00001A8A				1275+	DS	0H	
00001A8A	B96D3012			1276+	DC	XL4'B96D3012'	BDEPG r1,r2,R3
00001A8E	E310 5028 0024		00001A58	1278+	STG	R1,R1020	save R1 output
00001A94	07FB			1279+	BR	R11	return
00001A98				1280+RE20	DC	0F	xl16 expected result
00001A98				1281+	DROP	R5	
00001A98	00000000 00000000			1282	DC	XL8'0000 0000 0000 0000'	result
00001AA0	00000000 00000000			1283	DC	XL8'0000 0000 0000 0000'	R2
00001AA8	FFFFFFFF FFFFFFFF			1284	DC	XL8'FFFF FFFF FFFF FFFF'	R3
				1285			
				1286	RRF_A	BDEPG	
00001AB0				1287+	DS	0FD	
00001AB0		00001AB0		1288+	USING	*,R5	base for test data and test routine
00001AB0	00001AE8			1289+T21	DC	A(X21)	address of test routine
00001AB4	0015			1290+	DC	H'21'	test number

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00001AB6	00			1291+	DC	X'00'	
00001AB7	C2C4C5D7 C7404040			1292+	DC	CL8'BDEPG'	instruction name
00001AC0	00001B20			1293+	DC	A(RE21+8)	address of R2 source
00001AC4	00001B28			1294+	DC	A(RE21+16)	address of R3 source
00001AC8	00000008			1295+	DC	A(8)	result length
00001ACC	00001B18			1296+REA21	DC	A(RE21)	result address
00001AD0	00000000 00000000			1297+	DS	FD	gap
00001AD8	00000000 00000000			1298+R1021	DS	D	R1 output
00001AE0	00000000 00000000			1299+	DS	FD	gap
				1300+*			
00001AE8				1301+X21	DS	0F	
00001AE8	E310 8E88 0004		00001088	1302+	LG	R1,R1FUDGE	
00001AEE	B982 0022			1303+	XGR	R2,R2	
00001AF2	5820 5010		00000010	1304+	L	R2,R2ADDR	load r2 source address
00001AF6	E322 0000 0004		00000000	1305+	LG	R2,0(R2)	load source
00001AFC	B982 0033			1306+	XGR	R3,R3	
00001B00	5830 5014		00000014	1307+	L	R3,R3ADDR	load r3 source address
00001B04	E333 0000 0004		00000000	1308+	LG	R3,0(R3)	load source
00001B0A				1311+	DS	0H	
00001B0A	B96D3012			1312+	DC	XL4'B96D3012'	BDEPG r1,r2,R3
00001B0E	E310 5028 0024		00001AD8	1314+	STG	R1,R1021	save R1 output
00001B14	07FB			1315+	BR	R11	return
00001B18				1316+RE21	DC	0F	xl16 expected result
00001B18				1317+	DROP	R5	
00001B18	00010001 00010001			1318	DC	XL8'0001 0001 0001 0001'	result
00001B20	00010001 00010001			1319	DC	XL8'0001 0001 0001 0001'	R2
00001B28	FFFFFFFF FFFFFFFF			1320	DC	XL8'FFFF FFFF FFFF FFFF'	R3
				1321			
00001B30				1322	RRF_A	BDEPG	
00001B30		00001B30		1323+	DS	0FD	
00001B30	00001B68			1324+	USING	*,R5	base for test data and test routine
00001B34	0016			1325+T22	DC	A(X22)	address of test routine
00001B36	00			1326+	DC	H'22'	test number
00001B36	00			1327+	DC	X'00'	
00001B37	C2C4C5D7 C7404040			1328+	DC	CL8'BDEPG'	instruction name
00001B40	00001BA0			1329+	DC	A(RE22+8)	address of R2 source
00001B44	00001BA8			1330+	DC	A(RE22+16)	address of R3 source
00001B48	00000008			1331+	DC	A(8)	result length
00001B4C	00001B98			1332+REA22	DC	A(RE22)	result address
00001B50	00000000 00000000			1333+	DS	FD	gap
00001B58	00000000 00000000			1334+R1022	DS	D	R1 output
00001B60	00000000 00000000			1335+	DS	FD	gap
				1336+*			
00001B68				1337+X22	DS	0F	
00001B68	E310 8E88 0004		00001088	1338+	LG	R1,R1FUDGE	
00001B6E	B982 0022			1339+	XGR	R2,R2	
00001B72	5820 5010		00000010	1340+	L	R2,R2ADDR	load r2 source address
00001B76	E322 0000 0004		00000000	1341+	LG	R2,0(R2)	load source
00001B7C	B982 0033			1342+	XGR	R3,R3	
00001B80	5830 5014		00000014	1343+	L	R3,R3ADDR	load r3 source address
00001B84	E333 0000 0004		00000000	1344+	LG	R3,0(R3)	load source
00001B8A				1347+	DS	0H	
00001B8A	B96D3012			1348+	DC	XL4'B96D3012'	BDEPG r1,r2,R3

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
				1399	*****		
				1400	* Register equates		
				1401	*****		
		00000000	00000001	1403	R0	EQU	0
		00000001	00000001	1404	R1	EQU	1
		00000002	00000001	1405	R2	EQU	2
		00000003	00000001	1406	R3	EQU	3
		00000004	00000001	1407	R4	EQU	4
		00000005	00000001	1408	R5	EQU	5
		00000006	00000001	1409	R6	EQU	6
		00000007	00000001	1410	R7	EQU	7
		00000008	00000001	1411	R8	EQU	8
		00000009	00000001	1412	R9	EQU	9
		0000000A	00000001	1413	R10	EQU	10
		0000000B	00000001	1414	R11	EQU	11
		0000000C	00000001	1415	R12	EQU	12
		0000000D	00000001	1416	R13	EQU	13
		0000000E	00000001	1417	R14	EQU	14
		0000000F	00000001	1418	R15	EQU	15
				1420	*****		
				1421	* Register equates		
				1422	*****		
		00000000	00000001	1424	V0	EQU	0
		00000001	00000001	1425	V1	EQU	1
		00000002	00000001	1426	V2	EQU	2
		00000003	00000001	1427	V3	EQU	3
		00000004	00000001	1428	V4	EQU	4
		00000005	00000001	1429	V5	EQU	5
		00000006	00000001	1430	V6	EQU	6
		00000007	00000001	1431	V7	EQU	7
		00000008	00000001	1432	V8	EQU	8
		00000009	00000001	1433	V9	EQU	9
		0000000A	00000001	1434	V10	EQU	10
		0000000B	00000001	1435	V11	EQU	11
		0000000C	00000001	1436	V12	EQU	12
		0000000D	00000001	1437	V13	EQU	13
		0000000E	00000001	1438	V14	EQU	14
		0000000F	00000001	1439	V15	EQU	15
		00000010	00000001	1440	V16	EQU	16
		00000011	00000001	1441	V17	EQU	17
		00000012	00000001	1442	V18	EQU	18
		00000013	00000001	1443	V19	EQU	19
		00000014	00000001	1444	V20	EQU	20
		00000015	00000001	1445	V21	EQU	21

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES			
RE20	F	00001A98	4	1280	1257	1258	1260	
RE21	F	00001B18	4	1316	1293	1294	1296	
RE22	F	00001B98	4	1352	1329	1330	1332	
RE3	F	00001218	4	664	641	642	644	
RE4	F	00001298	4	700	677	678	680	
RE5	F	00001318	4	736	713	714	716	
RE6	F	00001398	4	772	749	750	752	
RE7	F	00001418	4	808	785	786	788	
RE8	F	00001498	4	844	821	822	824	
RE9	F	00001518	4	880	857	858	860	
REA1	A	000010CC	4	572				
REA10	A	0000154C	4	896				
REA11	A	000015CC	4	932				
REA12	A	0000164C	4	972				
REA13	A	000016CC	4	1008				
REA14	A	0000174C	4	1044				
REA15	A	000017CC	4	1080				
REA16	A	0000184C	4	1116				
REA17	A	000018CC	4	1152				
REA18	A	0000194C	4	1188				
REA19	A	000019CC	4	1224				
REA2	A	0000114C	4	608				
REA20	A	00001A4C	4	1260				
REA21	A	00001ACC	4	1296				
REA22	A	00001B4C	4	1332				
REA3	A	000011CC	4	644				
REA4	A	0000124C	4	680				
REA5	A	000012CC	4	716				
REA6	A	0000134C	4	752				
REA7	A	000013CC	4	788				
REA8	A	0000144C	4	824				
REA9	A	000014CC	4	860				
READDR	A	0000001C	4	455	262			
REG2LOW	U	000000DD	1	402				
REG2PATT	U	AABBCCDD	1	401				
RELEN	A	00000018	4	454				
RPTDWSAV	D	00000390	8	327	314	318		
RPTERROR	I	0000033C	4	300	275			
RPTSAVE	F	00000384	4	324	300	321		
RPTSVR5	F	00000388	4	325	301	320		
SKL0001	U	0000005E	1	220	236			
SKT0001	C	0000022A	20	217	220	237		
SVOLDPSW	U	00000140	0	156				
T1	A	000010B0	4	565	1370			
T10	A	00001530	4	889	1379			
T11	A	000015B0	4	925	1380			
T12	A	00001630	4	965	1381			
T13	A	000016B0	4	1001	1382			
T14	A	00001730	4	1037	1383			
T15	A	000017B0	4	1073	1384			
T16	A	00001830	4	1109	1385			
T17	A	000018B0	4	1145	1386			
T18	A	00001930	4	1181	1387			
T19	A	000019B0	4	1217	1388			
T2	A	00001130	4	601	1371			
T20	A	00001A30	4	1253	1389			

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES	
T21	A	00001AB0	4	1289	1390	
T22	A	00001B30	4	1325	1391	
T3	A	000011B0	4	637	1372	
T4	A	00001230	4	673	1373	
T5	A	000012B0	4	709	1374	
T6	A	00001330	4	745	1375	
T7	A	000013B0	4	781	1376	
T8	A	00001430	4	817	1377	
T9	A	000014B0	4	853	1378	
TESTING	F	00001004	4	413	256	
TNUM	H	00000004	2	448	255	303
TSUB	A	00000000	4	447	259	
TTABLE	F	00001BB8	4	1369		
V0	U	00000000	1	1424		
V1	U	00000001	1	1425		
V10	U	0000000A	1	1434		
V11	U	0000000B	1	1435		
V12	U	0000000C	1	1436		
V13	U	0000000D	1	1437		
V14	U	0000000E	1	1438		
V15	U	0000000F	1	1439		
V16	U	00000010	1	1440		
V17	U	00000011	1	1441		
V18	U	00000012	1	1442		
V19	U	00000013	1	1443		
V2	U	00000002	1	1426		
V20	U	00000014	1	1444		
V21	U	00000015	1	1445		
V22	U	00000016	1	1446		
V23	U	00000017	1	1447		
V24	U	00000018	1	1448		
V25	U	00000019	1	1449		
V26	U	0000001A	1	1450		
V27	U	0000001B	1	1451		
V28	U	0000001C	1	1452		
V29	U	0000001D	1	1453		
V3	U	00000003	1	1427		
V30	U	0000001E	1	1454		
V31	U	0000001F	1	1455		
V4	U	00000004	1	1428		
V5	U	00000005	1	1429		
V6	U	00000006	1	1430		
V7	U	00000007	1	1431		
V8	U	00000008	1	1432		
V9	U	00000009	1	1433		
X0001	U	000002B8	1	226	214	227
X1	F	000010E8	4	577	565	
X10	F	00001568	4	901	889	
X11	F	000015E8	4	937	925	
X12	F	00001668	4	977	965	
X13	F	000016E8	4	1013	1001	
X14	F	00001768	4	1049	1037	
X15	F	000017E8	4	1085	1073	
X16	F	00001868	4	1121	1109	
X17	F	000018E8	4	1157	1145	
X18	F	00001968	4	1193	1181	

DESC	SYMBOL	SIZE	POS	ADDR
------	--------	------	-----	------

Entry: 0

Image	IMAGE	7200	0000-1C1F	0000-1C1F
Region		7200	0000-1C1F	0000-1C1F
CSECT	MIE4TST	7200	0000-1C1F	0000-1C1F

STMT	FILE NAME
------	-----------

```
1 /devstor/sharedvfp/tests/mie4-01-BDXPG-BDEPG.asm
```

**** NO ERRORS FOUND ****